EFY TESTED
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Arduino Based **Liquid Vending Machine**

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This is an automatic liquid vending machine based on radio-frequency identification (RFID) system. A fixed amount of liquid can be dispensed by swiping an RFID tag across the RFID reader. An alphanumeric LCD is used to display the operation and instructions for users to follow while dispensing the liquid. This machine could be implemented in organisations like hospitals and colleges (medical, engineering, etc) to provide 24-hour service to customers with no humans involved.

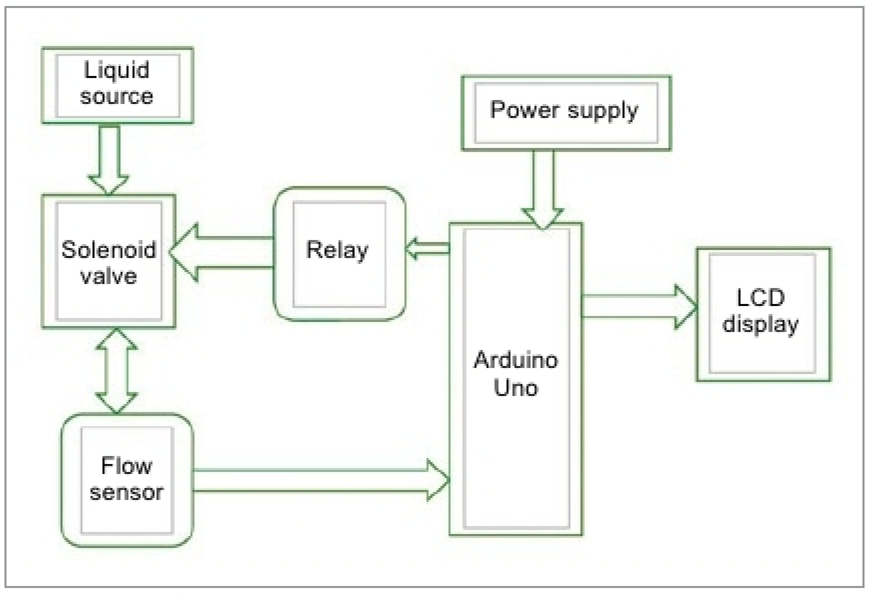

Fig. 1: Block diagram of Arduino based liquid vending machine

The vending machine has many advantages and is highly beneficial in many ways. Some examples are given below:

1. In hospitals, it can be used as a milk vending machine. Patients can buy milk from the vending machine without going to the market.

2. In educational organisations, some shops are opened for fixed hours. With this machine, you get 24-hour service.

3. At events/functions, this machine can be used to dispense juice of varied flavours.

4. In rural areas, this machine can be used to dispense water for a variety of purposes including drinking and watering plants to enhance productivity as well as quality.

The concept of this project could be useful for hobbyists and designers to further design a rugged-mechanical machine. This DIY article describes programming and interfacing the circuit as per the working prototype without mechanical construction parts. The main components required in this project are listed in the table on next page. The block diagram of Arduino based liquid vending machine is shown in Fig. 1.

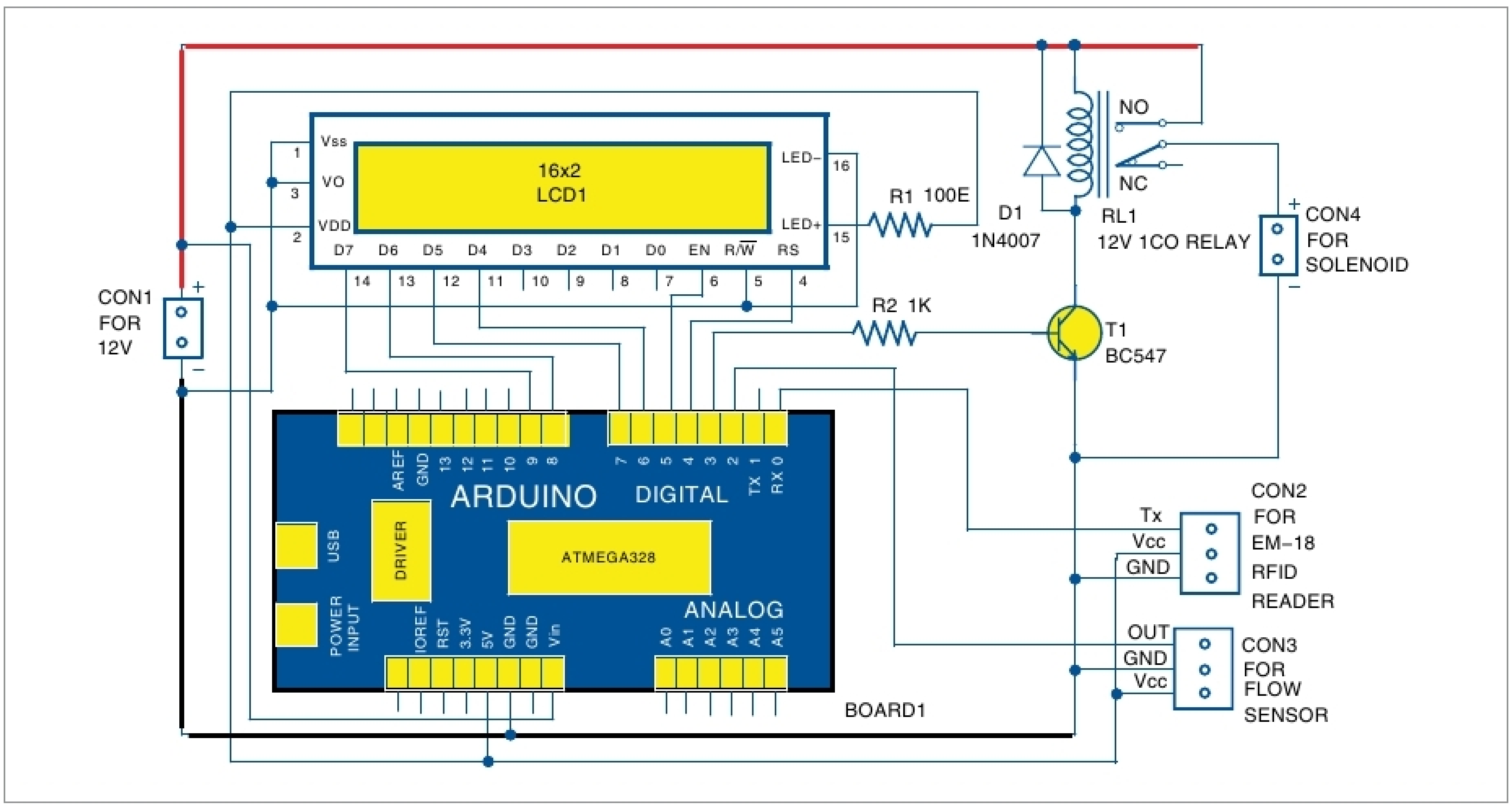

Fig. 2: Circuit diagram of the liquid vending machine